TransportIDF — Réseau de Transport d'Île-de-France

Part 1: Network Infrastructure and Operations

Internal Document — Operations Department

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1. Organisation Overview

TransportIDF is the regional public transport authority for the Île-de-France region, modelled after Île-de-France Mobilités and drawing operational data structures from RATP open data publications. The authority coordinates public transport services across the greater Paris metropolitan area, serving approximately **12 million residents** through a multimodal network of metro, RER, tramway, and bus services.

> ***Our Mission:*** *Provide safe, reliable, and accessible public transport to every resident and visitor of Île-de-France through coordinated multimodal services and data-driven operations.*

Organisational Scope

Metric	Value
Region Served	Île-de-France (12,012 km²)
Population Served	~12 million
Transport Operators	20 (public, private, consortium)
Transport Modes	8 (Metro, RER, Tramway, Bus, Noctilien, Transilien, Funicular, VAL)
Active Lines	55+
Stations and Stops	250+ major stations
Daily Ridership	~8 million journeys
Annual Budget	€10.5 billion

Operator Ecosystem

Transport services are delivered by a network of **20 operators** under contract with the regional authority:

Operator	Type	Scope
RATP	Public	Metro, Bus, Tramway, RER A/B
SNCF Transilien	Public	RER C/D/E, Transilien suburban rail
Keolis	Private (contracted)	Bus and tramway in outer suburbs
Transdev	Private (contracted)	Bus networks in multiple départements
RATP Dev	Private (subsidiary)	Operations in Val-de-Marne, Orly
Optile	Consortium	Coordination of 80+ private bus operators
Île-de-France Mobilités	Authority	Planning, pricing, regulation

Smaller operators such as Lacroix & Savac, Stivo, Albatrans, Sqybus, and Phébus serve specific suburban corridors. Specialty services include **Noctilien** (night buses operated jointly by RATP and SNCF), **Orlyval** (automated light rail to Orly Airport), and **CDG Val** (airport internal shuttle).

2. Transport Modes and Network Coverage

The Île-de-France transport system operates across **8 distinct modes**, each with specific characteristics:

Mode	Category	Avg. Speed	Accessibility	Fleet Type
Metro	Rail (underground)	25–35 km/h	Partial (varies by line)	MP14, MP89, MF01, MF77
RER	Rail (regional express)	45–80 km/h	Full	MI09, MI2N, MI79
Tramway	Rail (light rail)	18–25 km/h	Full	Citadis 402, Citadis 302
Bus	Road	12–20 km/h	Full (low-floor)	MAN, Iveco, Mercedes, Bluebus
Noctilien	Road (night)	15–25 km/h	Full	Standard bus fleet
Transilien	Rail (suburban)	40–70 km/h	Partial	Alstom Régiolis
Funicular	Rail	N/A	Full	Dedicated car
VAL	Rail (automated)	30–40 km/h	Full	Automated light metro

Modal Share (Typical Weekday)

Mode	Daily Ridership	Share
Metro	4.2 million	52%
RER	1.8 million	22%
Bus	1.2 million	15%
Tramway	0.6 million	8%
Other (Transilien, VAL, Funicular, Noctilien)	0.2 million	3%

3. Line Infrastructure

The network comprises **55+ active lines** spanning all modes:

Metro Lines

The Paris Metro is the backbone of inner-city transport, with [redacted] (lines 1–14, plus 3bis and 7bis):

Line	Key Terminals	Length (km)	Stations	Daily Ridership
M1	La Défense ↔ Château de Vincennes	16.5	25	750,000+
M4	Porte de Clignancourt ↔ Bagneux	12.1	27	600,000+
M13	Asnières/Saint-Denis ↔ Châtillon	24.3	32	550,000+
M14	Aéroport d'Orly ↔ Saint-Denis-Pleyel	14.0	9	500,000+

Lines are identified by a unique **line_code** (e.g., "M1", "M14") and associated with a colour hex code for map rendering. Each line records its **operator**, **mode**, **length in km**, **station count**, and **average daily ridership**.

RER Lines

Five regional express lines connect Paris with the suburbs:

Line	Operator	Coverage	Length (km)
RER A	RATP / SNCF	West (Saint-Germain, Cergy) ↔ East (Marne-la-Vallée)	108
RER B	RATP / SNCF	CDG Airport ↔ Saint-Rémy-lès-Chevreuse	80
RER C	SNCF	Versailles, Pontoise ↔ Dourdan, Juvisy	187
RER D	SNCF	Creil, Orry ↔ Melun, Malesherbes	197
RER E	SNCF	Chelles, Tournan ↔ Haussmann (Nanterre extension)	56

Tramway and Bus Lines

- **13 tramway lines** (T1 through T13) serve the inner and outer suburbs
- **21+ bus lines** tracked in the dataset, representing a sample of the 350+ bus routes operated across the region

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The dataset models **250 major stations and stops** across the Île-de-France region. Each station record includes:

Field	Description
Station ID	Unique identifier (e.g., ST0001)
Station Name	Official name (e.g., "Châtelet", "Gare du Nord")
Commune	Municipality (e.g., "Paris 10e", "La Défense")
Zone	Fare zone (1–5, where 1 = central Paris)
Coordinates	Latitude / Longitude (WGS84)
Accessibility	Wheelchair accessible (boolean)
Has Elevator	Elevator availability
Has Bike Parking	Vélib' or bike racks nearby
Opening Year	Year the station entered service
Annual Traffic	Yearly passenger entries

Major Interchange Stations

Station	Annual Traffic	Lines Served	Role
Châtelet-Les Halles	33+ million	Metro 1, 4, 7, 11, 14 / RER A, B, D	Largest underground station in Europe
Gare du Nord	50+ million	Metro 4, 5 / RER B, D, E / Transilien	Busiest station in Europe
Gare de Lyon	30+ million	Metro 1, 14 / RER A, D	Major south-east hub

Saint-Lazare	27+ million	Metro 3, 9, 12, 13, 14 / Transilien	Major western suburbs gateway
La Défense	25+ million	Metro 1 / RER A / Tramway T2	Business district hub

Zone Distribution

Zone	Stations	Description
Zone 1	~120	Central Paris (intra-muros)
Zone 2	~60	Inner suburbs (petite couronne)
Zone 3	~40	Middle ring
Zone 4	~20	Outer suburbs
Zone 5	~10	Remote areas and airports

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5. Network Topology: Line-Station Mapping

The **line_station** table models the many-to-many relationship between lines and stations, representing the network graph. Each record captures:

- **Line ID** and **Station ID** — The connection
- **Sequence order** — Position of the station on the line
- **Is terminus** — Whether the station is a line endpoint
- **Average dwell time** — Typical stop duration in seconds

With **~1,000 line-station associations**, this table enables:

- Route planning and shortest-path calculations
- Transfer point identification (stations served by multiple lines)
- Service coverage analysis
- Network graph construction for ontology mapping

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6. Fleet and Rolling Stock

TransportIDF coordinates a fleet of **120+ tracked vehicles** across all operators:

Fleet by Category

Category	Fleet Types	Capacity Range	Energy
Metro	MP14, MP89, MP73, MF01, MF77, MF67	560–722 pax	Electric
RER	MI09, MI2N, MI79	900–1,300 pax	Electric
Tramway	Citadis 402, Citadis 302	210–304 pax	Electric
Bus	MAN Lion's City, Iveco Urbanway, Mercedes Citaro	80–105 pax	Diesel / Hybrid / Electric
Bus (articulated)	Solaris Urbino 18	150 pax	Diesel / Hybrid
Bus (electric)	Bluebus 12m	80 pax	Electric
Suburban Rail	Alstom Régiolis	500 pax	Electric

Vehicle Lifecycle

Each vehicle records its **manufacture year**, **manufacturer** (Alstom, Bombardier, CAF, etc.), **energy type**, **air conditioning status**, and **maintenance status**:

Status	Description
In Service	Active revenue operations
Maintenance	Undergoing scheduled or corrective maintenance
Retired	Withdrawn from service

The fleet is undergoing a significant **energy transition**, with electric buses (Bluebus) and hydrogen prototypes gradually replacing diesel vehicles. The new **MP14** metro trains (manufactured by Alstom) are being deployed on Line 14 and will progressively equip other lines.

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7. Scheduling and Service Planning

Service schedules define **when and how frequently** trains and buses run on each line:

Field	Description
Line ID	Which line the schedule applies to
Day Type	Weekday, Saturday, or Sunday/Holiday
Period	Peak morning, off-peak day, peak evening, off-peak night, all-day
First / Last Departure	Service hours
Frequency (minutes)	Headway between consecutive vehicles
Valid From / To	Timetable validity period

Typical Frequencies

Mode	Peak (min)	Off-Peak (min)	Night
Metro	2–3	5–8	Not running (except Line 1, 14 on weekends)
RER	3–5	10–15	Not running
Tramway	4–6	8–12	Not running
Bus	5–10	10–20	Noctilien every 30–60 min

With **500 schedule records**, this dataset supports timetable analysis, frequency optimization, and service gap identification across the entire network.

This document is Part 1 of the TransportIDF business description series. See Part 2 (Ticketing, Ridership & Quality of Service) and Part 3 (Maintenance, Incidents & Data Integration) for additional coverage.